# Problem 20

Among students attending a probability theory class, we randomly select one. Let event $A$ mean that the selected student is male. Further, let event $B$ mean that he does not smoke, and event $C$ that he lives in a student dormitory.

1° Describe event $AB\bar{C}$.
2° Under what conditions does the identity $ABC = A$ hold?
3° When will the relation $\bar{C} \subseteq B$ be correct?
4° Explain when the equality $\bar{A} = B$ occurs, and does it occur when all present males do not smoke?

# Solution

## Given Events

- $A$ = selected student is male
- $B$ = selected student does not smoke
- $C$ = selected student lives in a student dormitory

Therefore:

- $\bar{A}$ = selected student is female
- $\bar{B}$ = selected student smokes
- $\bar{C}$ = selected student does not live in a dormitory

## Part 1°: Describe $AB\bar{C}$

### Analysis

The event $AB\bar{C}$ is the intersection of three events:

$$AB\bar{C} = A \cap B \cap \bar{C}$$

### Description

$$\boxed{AB\bar{C} = \text{The selected student is a male who does not smoke and does not live in a dormitory}}$$

This event occurs when all three conditions are satisfied simultaneously:

- The student is male (event $A$)
- The student does not smoke (event $B$)
- The student does NOT live in a dormitory (event $\bar{C}$)

## Part 2°: Under what conditions does $ABC = A$ hold?

### Analysis

The equation $ABC = A$ means:

$$A \cap B \cap C = A$$

This holds if and only if $A \subseteq B \cap C$, i.e., $A \subseteq B$ and $A \subseteq C$.

### Interpretation

$A \subseteq B$ means: Every male student does not smoke (all males are non-smokers)
$A \subseteq C$ means: Every male student lives in a dormitory (all males live in dorms)

### Answer

$$\boxed{ABC = A \iff \text{All male students are non-smokers AND all male students live in dormitories}}$$

Equivalently:

$$ABC = A \iff A \subseteq B \text{ and } A \subseteq C$$

**Condition:** Every male student must satisfy both conditions: not smoking and living in a dormitory.

### Verification

If all males are non-smokers and live in dorms, then: - Selecting a male ($A$) automatically means selecting a non-smoker ($B$) and dormitory resident ($C$) - Therefore $ABC = A$

Conversely, if $ABC = A$, then every element of $A$ must also be in $B$ and $C$, so $A \subseteq B \cap C$.

## Part $3^\circ$: When is $\bar{C} \subseteq B$ correct?

### Analysis

The relation $\bar{C} \subseteq B$ means:

Every student who doesn't live in a dormitory is a non-smoker

### Interpretation

This means: If a student does not live in a dormitory, then that student does not smoke.

Equivalently: All students living outside dormitories are non-smokers.

Or: No student who lives outside a dormitory smokes.

### Answer

$$\boxed{\bar{C} \subseteq B \iff \text{All students not living in dormitories are non-smokers}}$$

**Condition:** There are no smokers among students who live outside dormitories.

Equivalently: $\bar{C}\bar{B} = \emptyset$ (no one both lives outside dorms AND smokes)

Or using the contrapositive: $\bar{B} \subseteq C$ (all smokers live in dormitories)

## Part $4^\circ$: When does $\bar{A} = B$ occur? Does it occur when all males don't smoke?

### Part (a): When does $\bar{A} = B$ hold?

The equality $\bar{A} = B$ means:

The set of female students equals the set of non-smoking students

This requires two conditions:

1. $\bar{A} \subseteq B$: All females are non-smokers
2. $B \subseteq \bar{A}$: All non-smokers are female

**Combined meaning:**

$$\boxed{\bar{A} = B \iff \text{A student is female if and only if they don't smoke}}$$

Equivalently:

- All females are non-smokers
- All males are smokers
- All non-smokers are female
- All smokers are male

**Part (b): Does $\bar{A} = B$ occur when all males don't smoke?**

**Given condition:** All present males do not smoke, i.e., $A \subseteq B$.

**Question:** Does this imply $\bar{A} = B$?

**Analysis:**

If all males don't smoke ($A \subseteq B$), this means:

$$A \cap \bar{B} = \emptyset \quad \text{(no male smokers)}$$

For $\bar{A} = B$ to hold, we need:

- All females don't smoke: $\bar{A} \subseteq B$
- All non-smokers are female: $B \subseteq \bar{A}$

But if all males don't smoke ($A \subseteq B$), then males are among the non-smokers, so $B \nsubseteq \bar{A}$.

Therefore:

$$\boxed{\text{NO, } \bar{A} = B \text{ does NOT occur when all males don't smoke}}$$

**Reason:** When all males don't smoke, the non-smoking group ($B$) includes males, so $B \neq \bar{A}$ (which contains only females).

For $\bar{A} = B$ to hold, we would need:

- All males smoke (so no males in $B$)
- All females don't smoke (so all of $\bar{A}$ is in $B$)

This is the **opposite** of the given condition.

**Counterexample**

Suppose:

- There are 10 males, all non-smokers
- There are 5 females, all non-smokers

Then:

- $\bar{A} = 5$ females
- $B = 15$ non-smokers (10 males + 5 females)
- $\bar{A} \neq B$ since $|\bar{A}| = 5 \neq 15 = |B|$

# Final Answer

$$\boxed{\begin{array}{ll} 1^\circ & AB\bar{C} = \text{Male, non-smoker, not living in dormitory} \\ 2^\circ & ABC = A \iff \text{All males are non-smokers and live in dormitories} \\ 3^\circ & \bar{C} \subseteq B \iff \text{All students not in dorms are non-smokers} \\ 4^\circ & \bar{A} = B \iff \text{Student is female iff non-smoker} \\ & \text{NO, this does NOT occur when all males don't smoke} \end{array}}$$

**Key insight for $4^\circ$:** The equality $\bar{A} = B$ requires that females and non-smokers are exactly the same set. This would require all males to smoke (not don't smoke) and all females to not smoke.